
Enhancing Few-Shot Robotic Manipulation by Replacing Depth Sensors with Learned 3D Geometry from MAST3R

Ruben Figge
University of Amsterdam

Sjoerd Gunneweg
University of Amsterdam

Jurgen de Heus
University of Amsterdam

Mees Lindeman
University of Amsterdam

Abstract

Precision in robotic manipulation typically requires expensive RGB-D sensors for 3D spatial understanding, making large-scale deployments difficult. This work investigates if MAST3R, a method for 3D reconstruction from RGB images alone, is able to effectively replace depth sensors in robotic manipulation pipelines. We present an integration of MAST3R with RVT-2 (Robotic Vision Transformer), doing so creates an RGB-only manipulation system that allows for depth estimation without dedicated hardware. Our approach consists of three steps: reproducing MAST3R results on the DTU benchmark, evaluating MAST3R-generated depth maps on RLBench manipulation tasks, and extending RVT-2 to accept the learned 3D geometry as input. While we encountered reproducibility challenges with the original MAST3R implementation, achieving 0.468 overall Chamfer distance compared to the reported 0.374 on DTU, we successfully demonstrated the technical feasibility of RGB-only manipulation. Our comprehensive evaluation across four manipulation tasks (`close_jar`, `open_drawer`, `insert_onto_square_peg`, `push_buttons`) reveals significant view-dependent variations in depth estimation quality, with SSIM scores of 0.6-0.8 indicating good structural preservation despite local depth inaccuracies. The integration of MAST3R in RVT-2 achieved stable training convergence with minimal code modifications. However, evaluation of the trained models remains an open issue which should be addressed in future work. Nevertheless, this work sets the foundation for more accessible robotic manipulation systems by demonstrating that learned 3D geometry can potentially replace expensive sensing hardware, while highlighting key technical challenges that must be addressed for practical deployment.

1 Introduction

Precision in robotic manipulation requires accurate 3D understanding. However, to achieve this, state-of-the-art models like RVT-2 (Robotic Vision Transformer) [Goyal et al., 2024] depend on costly RGB-D sensors. Recent advances in 3D reconstruction offer an alternative: MAST3R (Matching And Stereo 3D Reconstruction) [Leroy et al., 2024], an augmentation of the DUST3R [Wang et al., 2024] framework, can produce 3D reconstructions from RGB images alone, without requiring depth sensors or camera calibration. The integration of a method like MAST3R could potentially provide sufficiently accurate results for environmental interpretation in scenarios where depth sensors are unavailable, potentially reducing the need for extensive calibration procedures. This work investigates whether depth maps generated with the use of MAST3R can replace RGB-D inputs in robotic manipulation pipelines. We integrate MAST3R with RVT-2 in a three-step approach: reproducing MAST3R results

on DTU [Aanæs et al., 2016], benchmarking depth maps generated using MAST3R on RLBench [James et al., 2020] in a robotics context, and extending RVT-2 with the depth maps to create an RGB-only manipulation pipeline.

2 Related Work

2.1 3D Reconstruction from Uncalibrated Images

Recovering 3D structure from uncalibrated image sequences is a foundational problem in computer vision with applications in SLAM, localization, and robotics. Classical multi-view stereo (MVS) approaches follow a three-step pipeline: keypoint detection, descriptor extraction, and pairwise feature matching. Techniques like SIFT [Lowe, 2004] remain influential, but degrade under wide viewpoint changes, textureless surfaces, or repetitive patterns. Recent learning-based systems, such as DUST3R [Wang et al., 2024], address these limitations by treating image correspondence as a 3D prediction task. Instead of relying on sparse matching between image pairs, DUST3R jointly infers correspondences and 3D geometry through a transformer-based pipeline, significantly improving robustness. MAST3R [Leroy et al., 2024] builds upon DUST3R by introducing dense local features via a modified ViT backbone, enabling fine-grained depth prediction and dense stereo reconstruction. This results in state-of-the-art performance on Map-Free Localization [Arnold et al., 2022] and DTU [Aanæs et al., 2016], with particular robustness under large baselines and low-texture scenes.

2.2 Vision-Based Robotic Manipulation

High-precision robotic manipulation requires accurate 3D spatial perception, typically obtained using calibrated RGB-D sensors. Recent work such as RVT-2 [Goyal et al., 2024] reduces the supervision burden for manipulation via a transformer-based architecture that fuses multi-view features and predicts keypoints in 3D space. RVT-2 achieves strong results with few demonstrations on tasks from RLBench [James et al., 2020], a standard manipulation benchmark. However, it still assumes access to high-quality depth information. Bridging the gap between visual 3D reconstruction and robotic action prediction remains underexplored. While prior work has explored depth fusion or SLAM-based integration for perception, few systems have attempted to replace depth sensors entirely using learned geometry from raw RGB. This is particularly relevant in settings where depth sensors are impractical due to lighting, range, or surface properties.

3 Reproducibility of the Current Work

3.1 Description of the Method and Experiments

To validate MAST3R’s [Leroy et al., 2024] capabilities for downstream robotic tasks, we first reproduced the MVS results presented in the original paper, utilizing the official implementation and pretrained checkpoints. We focused exclusively on reproducing the MVS results due to limited time and the direct relevance of MVS to our project.

For this evaluation, we used the DTU dataset [Aanæs et al., 2016], a standard benchmark for multi-view stereo. DTU includes 124 indoor scenes, each captured from either 49 or 64 calibrated viewpoints under controlled lighting conditions. MAST3R performs MVS by triangulating matches obtained between pairs of views. Matching is conducted in a zero-shot manner without prior knowledge of camera parameters; camera poses are only utilized later for triangulating the matches into the ground truth reference frame. Spurious 3D points are subsequently filtered out using geometric consistency post-processing. Finally, the predicted point clouds are evaluated using average accuracy, completeness, and Chamfer distance metrics as specified by the benchmark authors.

3.2 Discussion and Analysis

Reproducing the MVS results of MAST3R proved challenging, as the GitHub repository lacks evaluation scripts, explicit support for the DTU dataset, and a clear pipeline for performing MVS through triangulation. These omissions left several key details unclear. Despite contacting the authors for clarification, we did not receive a response. Nonetheless, we closely adhered to the original paper’s

description and evaluated MAST3R accordingly. As shown in Table 1, our reproduction yielded worse results compared to those originally reported by MAST3R. Although we carefully followed the available instructions, this discrepancy could be due to undisclosed differences in procedure. Given the absence of detailed information, we conclude that MAST3R’s reported results for MVS are not entirely reproducible. However, the authors’ claim of achieving reasonable results in a zero-shot setting remains valid. An example of qualitative results can be seen in Figure 2 in Appendix A.1.

Methods	Acc.↓	Comp.↓	Overall↓
DUST3R (Original Results) Wang et al. [2024]	2.677	0.805	1.741
MASt3R (Original Results) Leroy et al. [2024]	0.403	0.344	0.374
MASt3R (Our Reproduction)	0.551	0.385	0.468

Table 1: Accuracy (Acc.), Completeness (Comp.), and Overall (Chamfer distance) scores for MAST3R evaluated on the DTU dataset Aanæs et al. [2016]. Lower is better for all metrics. Original results compared with our reproduced results.

An important consideration is that MVS evaluation via triangulation may not accurately reflect real-world performance if the ultimate goal is 3D reconstruction from uncalibrated images. Triangulation relying on predicted camera parameters typically produces poorer results compared to other methods utilizing predicted camera information. The MVS results reported by DUST3R [Wang et al., 2024], also reported in Table 1, were obtained through global alignment of predicted 3D points using known ground truth camera parameters. This approach works effectively even without initial camera knowledge, as the predicted camera parameters and 3D points are jointly optimized during alignment. Future work could involve comparing MAST3R’s MVS results, obtained via global alignment, with those of DUST3R to determine which method better supports downstream tasks involving uncalibrated images. This comparison was not included in this report, as the implementation of MAST3R’s global alignment using ground truth camera data is currently incomplete.

4 Extension

To explore whether MAST3R can support robotic manipulation, we extend its use beyond traditional MVS datasets by applying it to robot-centric scenes. This allows us to assess its suitability as a depth estimator backbone for RGB-only manipulation pipelines like RVT-2.

4.1 Description of the Method and Experiments

Benchmarking MAST3R on depth estimation of RL Bench. The first step of the extensions is to replace the ground-truth depth information provided by the RL Bench [James et al., 2020] dataset with depth estimations that are obtained with the use of MAST3R. To get more insights into the results of RVT-2 when using the estimated depth maps, the process is dissected by doing an analysis on the quality of the estimated depth maps by MAST3R. The results and further information on this analysis can be found in Appendix B.

The depth maps are created by processing all image pairs from each scene with MAST3R, combining the resulting point clouds through global alignment. After this depth maps are extracted from the unified point cloud. The method for global alignment being used is from DUST3R, the predecessor of MAST3R. RL Bench RGB images (128 x 128) are upsampled to 256 x 256 using bicubic interpolation to meet MAST3R’s input specifications. Due to errors occurring during global alignment because of high view pair variance, we reduced triangulation iterations from 300 to 150. Timesteps that kept giving errors are excluded from benchmarking and replaced with ground-truth depth maps for the training of RVT-2. Details about these timesteps can be found in Appendix C.

MASt3R to RVT-2 Integration. We modified the RVT-2 pipeline to accept MAST3R-generated point maps as a drop-in replacement for the original depth-derived point clouds. This required minimal changes, only a few lines of code were needed to adapt the PerAct data loading utilities to accept MAST3R-generated depth maps. To evaluate this RGB-only setup, we trained RVT-2 on four RL Bench tasks: `close_jar`, `open_drawer`, `insert_onto_square_peg`, and `push_buttons`. These were chosen to capture a range of difficulty and expected variance in performance, while also

considering storage constraints as MAST3R roughly doubles the data footprint per episode. Task selection was further limited by available SBU quota and compute resources.

4.2 Discussion and Analysis

Running RVT-2 was technically challenging due to outdated dependencies and compatibility issues. Despite the documentation, the model still relied on PyTorch3D [Ravi et al., 2020], which was fragile due to CUDA version sensitivity. CoppeliaSim’s [Rohmer et al., 2013] headless mode required Xvfb, whose integration demanded extensive debugging. These, and other infrastructure hurdles consumed substantial time and limited deeper model exploration, as well as cross seed analysis.

Due to current limitations in the evaluation pipeline, our analysis is based solely on training loss curves. Specifically, RVT-2 relies on CoppeliaSim to generate depth maps internally during evaluation, which makes it incompatible with externally provided MAST3R-generated point maps at test time. This introduces a domain gap between the depth data used during training and that used during evaluation. Despite this, average loss curves show that RVT-2 trained on MAST3R-generated point maps learns comparably to the original RGB-D baseline, indicating that the model is capable of learning meaningful behavior from MAST3R input. However, the relative loss results reveal task-specific differences: for simpler tasks like `close_jar`, MAST3R performs on par with or better than the baseline, while more precision-sensitive tasks such as `insert_peg` and `open_drawer` exhibit higher loss, particularly in the grip-related component. It’s important to note, however, that in robotic manipulation, training loss does not necessarily translate to higher task success. Loss values may not fully capture the model’s ability to execute precise physical interactions. As such, while these curves provide useful insight into training behavior and model sensitivity to input quality, they should not be interpreted as direct indicators of real-world performance.

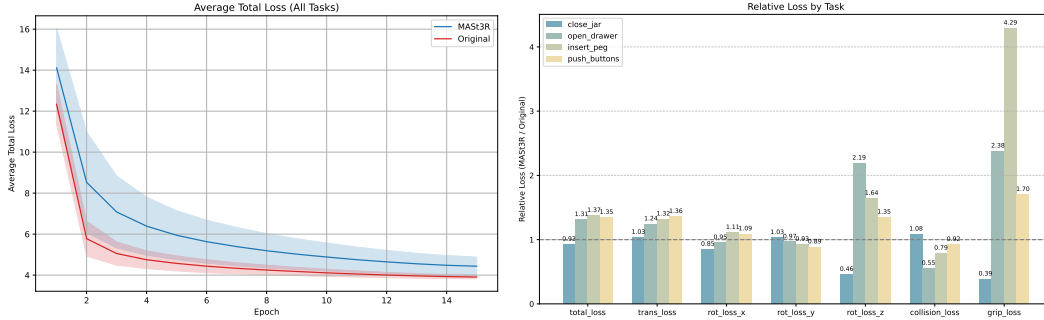


Figure 1: Comparison of average and relative loss during training for RVT-2 with MAST3R vs. Original RGB-D input

5 Conclusion & Future Work

This project explored the integration of MAST3R-generated 3D point maps into RVT-2 to replace conventional RGB-D inputs for high-precision robotic manipulation. Our results show that RVT-2 trained on MAST3R input achieves comparable average total loss to the original setup, indicating that the overall pipeline is viable. For a detailed view of task-specific performance, all loss curves are provided in Appendix D. Task-specific analysis revealed notable degradation in grip-related performance. A possible explanation is that MAST3R’s depth predictions were especially poor for the wrist camera view, highly linked to grasp evaluation, leading to increased grip loss. While the approach shows promise, future work should aim to improve reconstruction quality at critical viewpoints where depth errors significantly impact gripper performance. Additionally, future extensions could explore feeding MAST3R’s dense feature maps directly into RVT-2’s transformer backbone as an alternative to point maps.

References

- Henrik Aanaes, Rasmus Ramsbøl Jensen, George Vogiatzis, Engin Tola, and Anders Bjarholm Dahl. Large-scale data for multiple-view stereopsis. *International Journal of Computer Vision*, 120: 153–168, 2016.
- Eduardo Arnold, Jamie Wynn, Sara Vicente, Guillermo Garcia-Hernando, Áron Monszpart, Victor Prisacariu, Daniyar Turmukhambetov, and Eric Brachmann. Map-free visual relocalization: Metric pose relative to a single image. In Shai Avidan, Gabriel Brostow, Moustapha Cissé, Giovanni Maria Farinella, and Tal Hassner, editors, *Computer Vision – ECCV 2022*, pages 690–708, Cham, 2022. Springer Nature Switzerland. ISBN 978-3-031-19769-7.
- David Eigen, Christian Puhersch, and Rob Fergus. Depth map prediction from a single image using a multi-scale deep network. *CoRR*, abs/1406.2283, 2014. URL <http://arxiv.org/abs/1406.2283>.
- Ankit Goyal, Valts Blukis, Jie Xu, Yijie Guo, Yu-Wei Chao, and Dieter Fox. Rvt-2: Learning precise manipulation from few demonstrations. *arXiv preprint arXiv:2406.08545*, 2024.
- Stephen James, Zicong Ma, David Rovick Arrojo, and Andrew J Davison. Rlbench: The robot learning benchmark & learning environment. *IEEE Robotics and Automation Letters*, 5(2):3019–3026, 2020.
- Vincent Leroy, Yohann Cabon, and Jérôme Revaud. Grounding image matching in 3d with mast3r. In *European Conference on Computer Vision*, pages 71–91. Springer, 2024.
- David G Lowe. Distinctive image features from scale-invariant keypoints. *International journal of computer vision*, 60:91–110, 2004.
- Nikhila Ravi, Jeremy Reizenstein, David Novotny, Taylor Gordon, Wan-Yen Lo, Justin Johnson, and Georgia Gkioxari. Accelerating 3d deep learning with pytorch3d. *arXiv:2007.08501*, 2020.
- E. Rohmer, S. P. N. Singh, and M. Freese. Coppeliassim (formerly v-rep): a versatile and scalable robot simulation framework. In *Proc. of The International Conference on Intelligent Robots and Systems (IROS)*, 2013. www.coppeliarobotics.com.
- Shuzhe Wang, Vincent Leroy, Yohann Cabon, Boris Chidlovskii, and Jerome Revaud. Dust3r: Geometric 3d vision made easy. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 20697–20709, 2024.

A Reproduction

A.1 Qualitative MVS Results on DTU



Figure 2: Qualitative reproduction MVS results on DTU.

In Figure 2 we show some of the qualitative results of reproducing MAST3R on DTU. These point clouds are obtained by removing extreme outliers based on the distance to the median.

A.2 RVT-2 Reproduction Results

Although our primary goal was not to reproduce RVT-2, we first needed to verify that the provided codebase functioned correctly with our MAST3R-generated depth maps and could run reliably on the Snellius HPC-cluster. We successfully reproduced RVT-2 without modifying its model architecture or training logic. To validate our setup, we ran both the pretrained model and new training runs on two benchmark RLBench tasks: `close_jar` and `open_drawer`. As shown in Table 2, the results matched, considering the variance, that which was reported in the original paper, confirming that our setup was correct and suitable for running the MAST3R-based integration.

Table 2: Evaluation accuracy (%).

Task	Paper	Model	Trained
open_drawer	74.0 $\pm$ 11.8	88.0	72.0
close_jar	100.0 $\pm$ 0.0	100.0	100.0

B Depth Map Evaluation

To evaluate the quality of MAST3R-generated depth maps on RLBench scenes, we adopted standard metrics that assess both pixel-wise accuracy and perceptual structure. Specifically, we computed Absolute Relative Error (AbsRel), Root Mean Squared Error (RMSE), and Scale-Invariant RMSE to measure numerical accuracy, with lower values indicating better performance. We also report the Structural Similarity Index Measure (SSIM) [Eigen et al., 2014], which ranges from 0 to 1 and captures image-level structural fidelity. Results are reported across selected scenes using comparative plots.

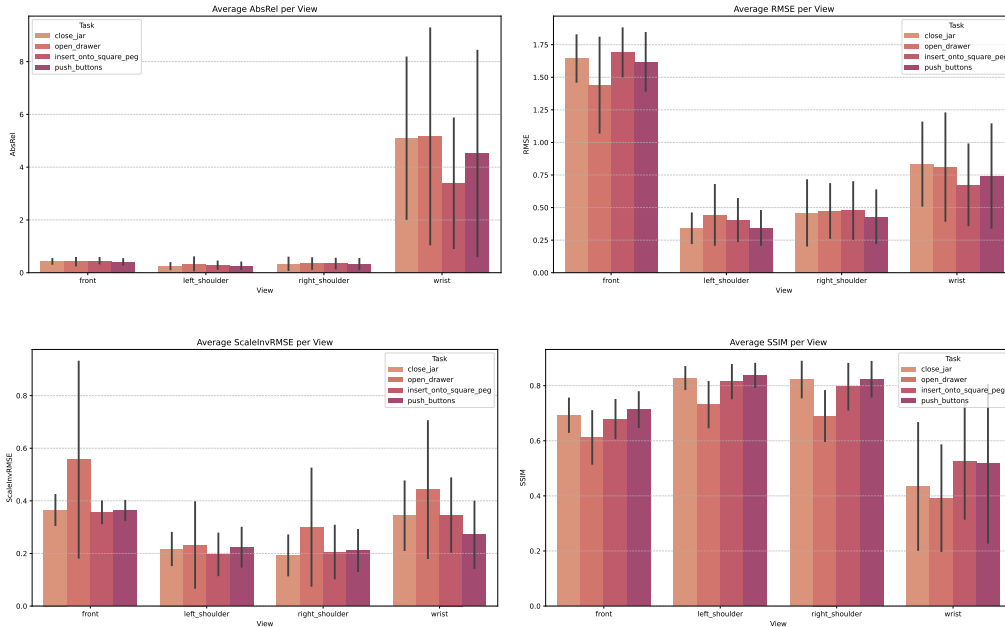


Figure 3: Depth reconstruction quality by camera view

We observe clear differences in depth estimation quality across camera views. The wrist view shows high AbsRel error, likely due to its limited field of view and the frequent overestimation of depth, as seen in the qualitative depth maps. These overpredictions lead to large relative errors, especially for nearby objects. Interestingly, RMSE is highest for the front view, despite its strong performance on other metrics. While foreground object contours are accurately reconstructed, the background is predicted to be significantly closer than ground truth, resulting in large squared errors. Finally, SSIM

is relatively high (0.6–0.8) across all views, indicating that the structural layout of the scenes is well preserved. This suggests that MAST3R captures global geometry coherently, even when depth values are locally inaccurate.

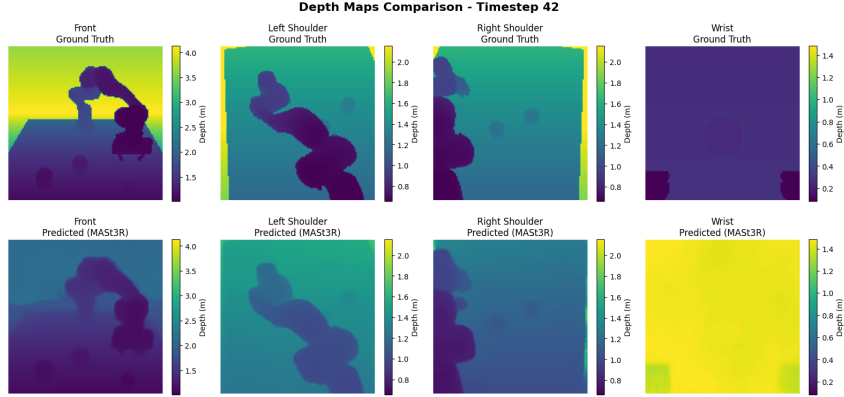


Figure 4: Qualitative results of depth estimation with the use of MAST3R. The scene depicted is timestep 42, episode 42 of the `close_jar` training observation.

C Scenes Failing Global Alignment

During the generation of the depth maps some scenes gave the following error: *"OpenCV(4.11.0) /io/opencv/modules/calib3d/src/sqnpn.cpp:253: error: (-215:Assertion failed) point_coordinate_variance >= POINT_VARIANCE_THRESHOLD in function 'computeOmega'"*.

The error originates in OpenCV’s PnP (Perspective-n-Point) implementation. It indicates that the 3D input points used for pose estimation do not have enough spatial variation. This could be because they’re too close together or lie nearly on a plane, which makes it mathematically unstable or ambiguous to determine a reliable camera pose. The number of scenes that had this error are displayed in Table 3.

Table 3: Number of scenes where the global alignment failed by task and dataset split.

Task	Train	Validation
push_buttons	47	18
insert_onto_square_peg	119	24
close_jar	93	16
open_drawer	5	2

During the benchmarking process, scenes with faulty depth maps are excluded from evaluation. However, for the training of RVT-2 [Goyal et al., 2024], we adopt a different strategy to preserve training data integrity. When estimated depth maps are unavailable due to global alignment errors, we replace them with ground-truth depth maps from the dataset.

This approach is necessary because a single faulty depth map would compromise an entire episode, despite all other timesteps being processed correctly. Getting rid of such episodes would result in the loss of nearly all training episodes for certain tasks, severely limiting the model’s learning capacity.

To address this issue, we intervene by replacing faulty estimated depth maps with ground-truth alternatives. This strategy maintains training continuity while minimizing impact on the overall learning process. The impact remains minimal since these replacements affect only a small number of scenes per episode. This preserves the realistic training conditions for the vast majority of the dataset.

D Per-Task Training Loss Curves

We present total loss curves for each individual task, comparing RVT-2 trained with MAST3R-inferred point maps against the original RGB-D input. These plots provide a more granular view of the training behavior and model sensitivity across different manipulation scenarios.

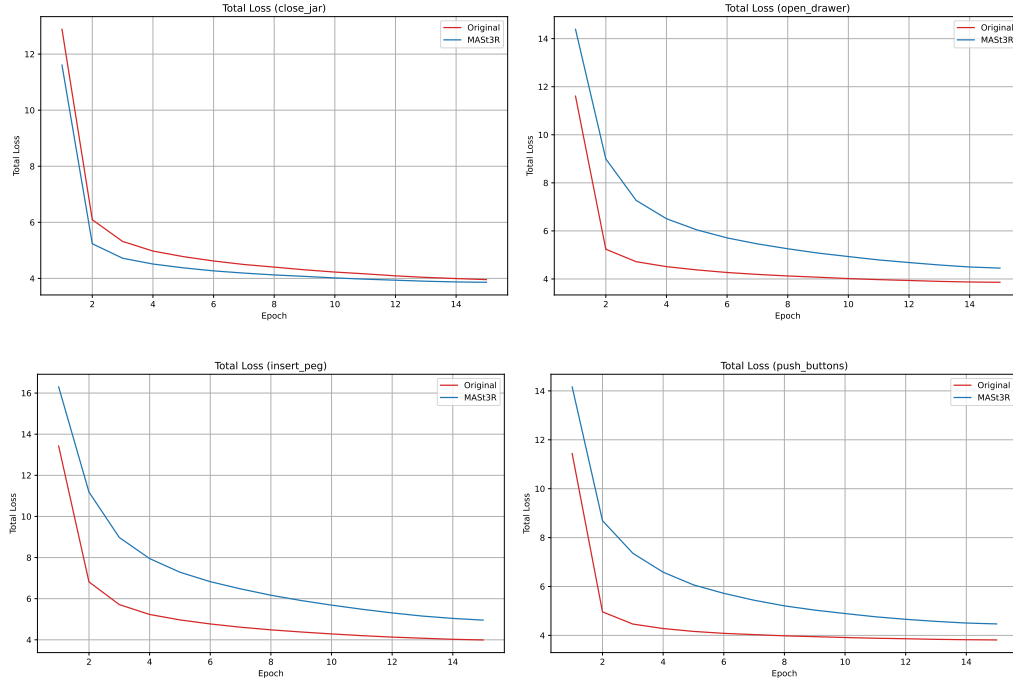


Figure 5: Per-task total training loss over epochs: MAST3R vs. Original RGB-D input